GPU-accelerated storage with SCADA

Taming AI Storage Systems

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What participants will learn

01 How GPU-initiated requests differ from direct DMA

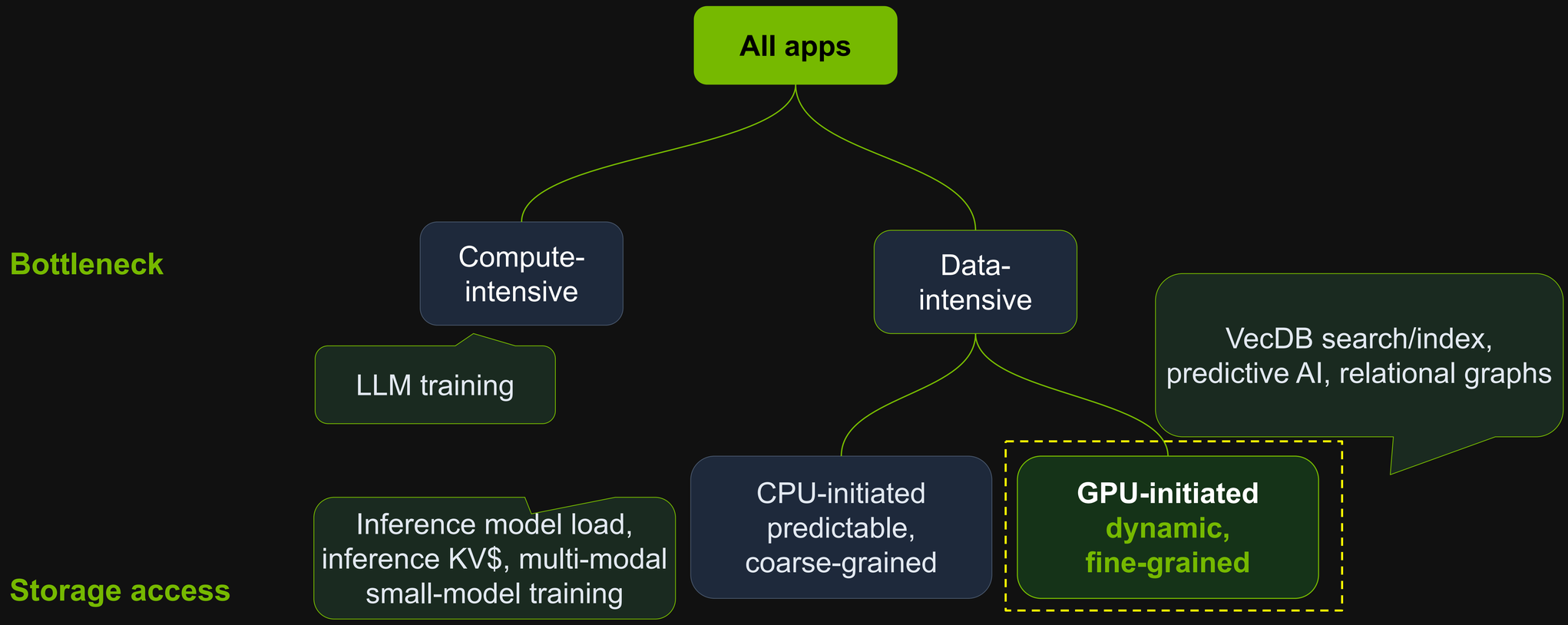
02 Why fine-grained, irregular reads benefit from concurrency

03 How topology and cache behavior affect storage access

04 How Nsight Systems helps locate a performance constraint

App taxonomy: bottlenecks, dynamism, granularity

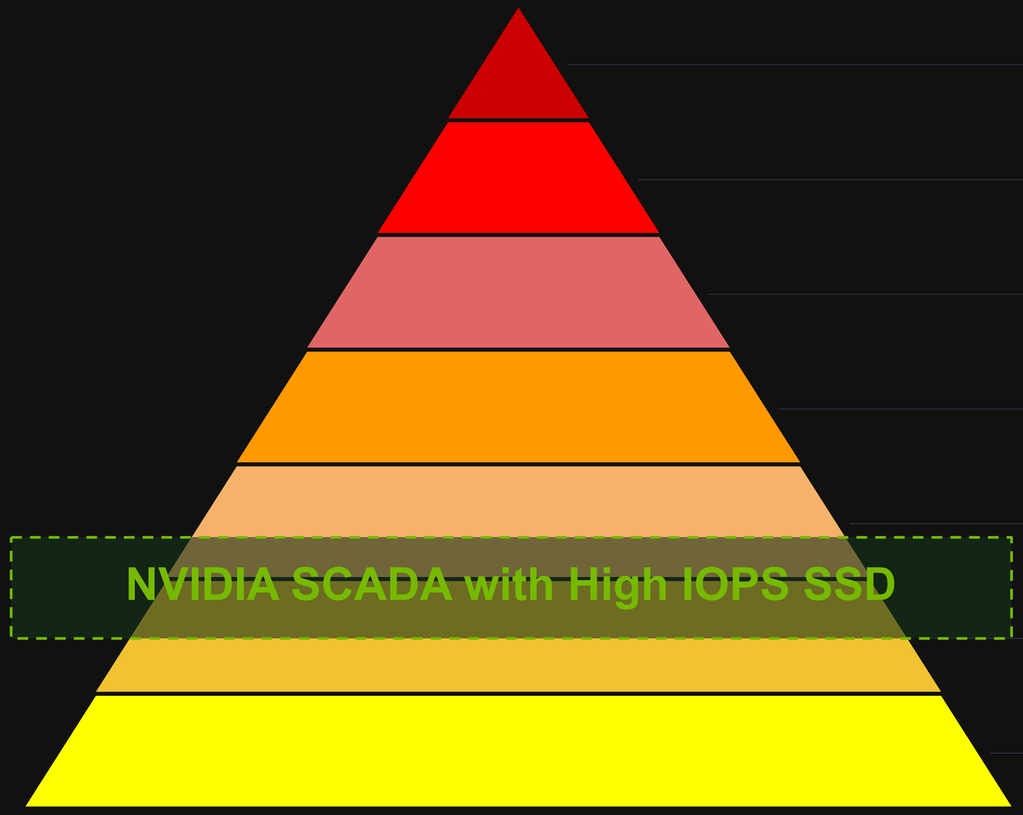
Designing storage solutions hinges on understanding app requirements



- Traditional focus for GPUs has been compute-intensive apps like training
- The explosion of innovation for VecDB, predictive AI drives new technologies to fill the gaps

GPU Memory Hierarchy and SCADA Place

Faster access · smaller capacity · higher cost per bit



GPU registers
Per-thread working values

GPU L1 / shared memory
On-chip SRAM · per SM

GPU L2 cache
On-chip SRAM · GPU-wide

GPU HBM
Device DRAM · G1

System RAM · DDR5
Host DRAM · G2

Local NVMe SSD
NAND flash media · G3

Shared Storage
Network Storage (RoCE, Infiniband) · G4

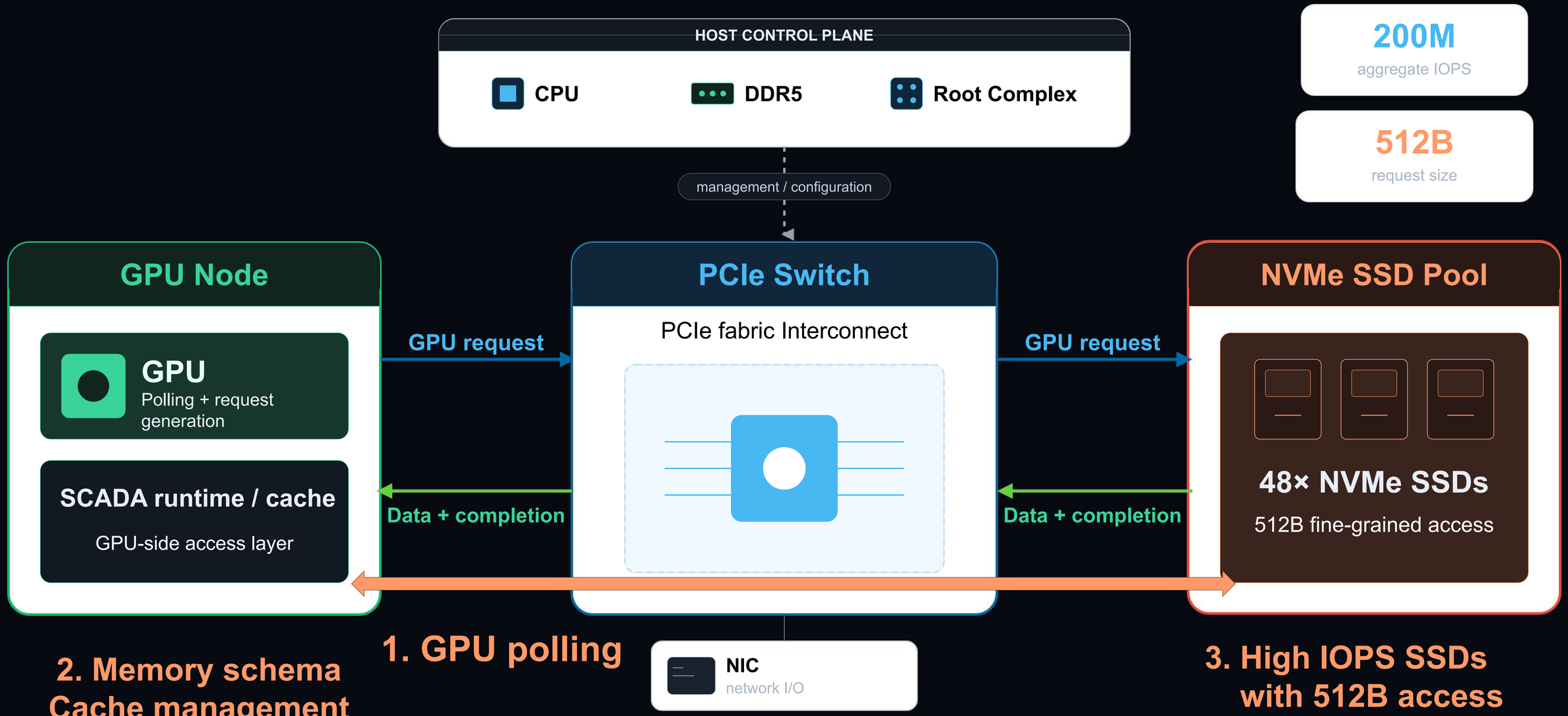
Memory

Storage

Slower access · larger capacity · lower cost per bit

Conceptual ordering; sizes are not to scale. G4 shared storage may use SSD, HDD, or other media.

Storage-Next Converged -- 200M IOPS (5ns per request)

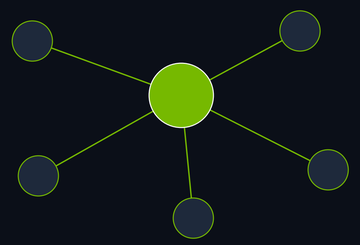


Suitable Applications for NVIDIA SCADA

Best fit: large datasets beyond GPU memory + many concurrent, fine-grained GPU reads

GNN training

Strong architectural fit



Sample neighbors → fetch features

Graph features beyond HBM

Scale training datasets; reduce feature-fetch stalls

Recommendation / GNN inference

Strong architectural fit



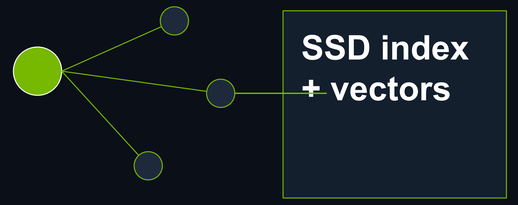
Embeddings

Sparse embedding / feature reads

Recommend at scale; detect fraud with large feature tables

Vector databases

Storage integration target



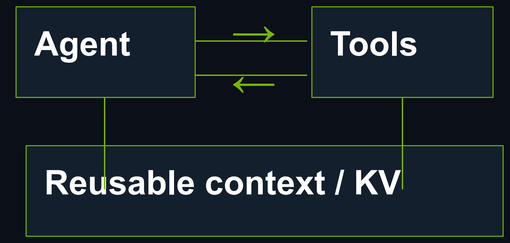
ANN candidates → on-demand reads

Disk-backed ANN search & RAG

Fetch candidate vectors and index data as needed

LLM & Agentic AI

KV integration to evaluate



Context reuse across agent steps

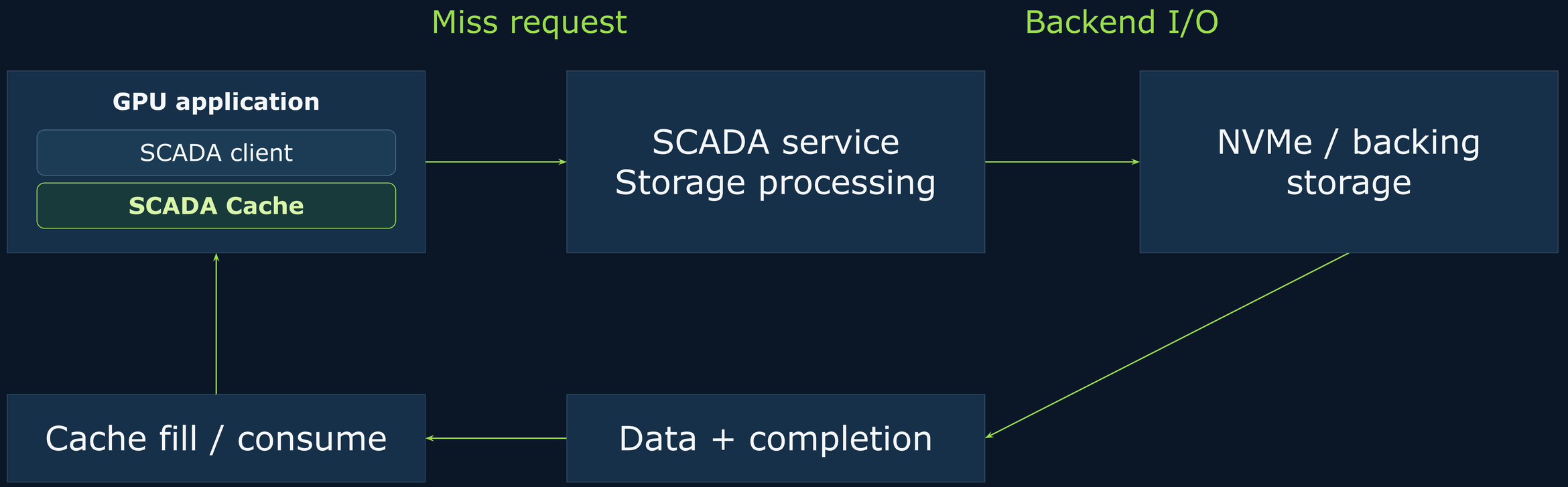
Warm KV and context reuse

Reuse KV to avoid prefill; benefit depends on transfer cost

GPU application → SCADA data access → local or remote storage

Fit assessment, not measured gains. Application engines own graph sampling, search, and KV management.

SCADA request, data, and completion flow



Cache hits stay in the GPU client. This is a logical read path.

System1 for NVIDIA Storage-Next

44 x E1.S SSDs



2 x
PCIe Gen 6
Switches

ConnectX-8



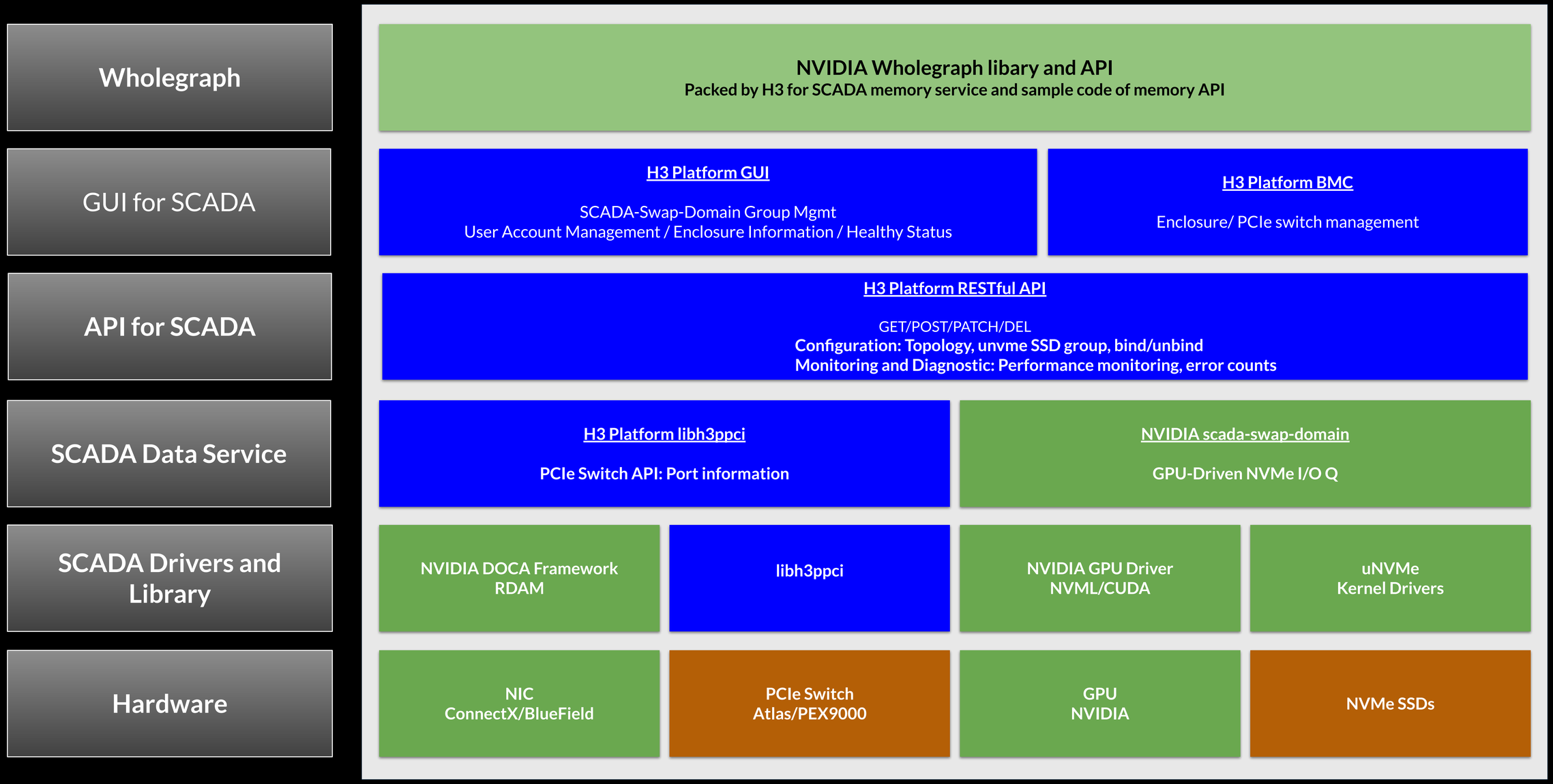
RTX Pro
6000

H3 System2 for NVIDIA Storage-Next



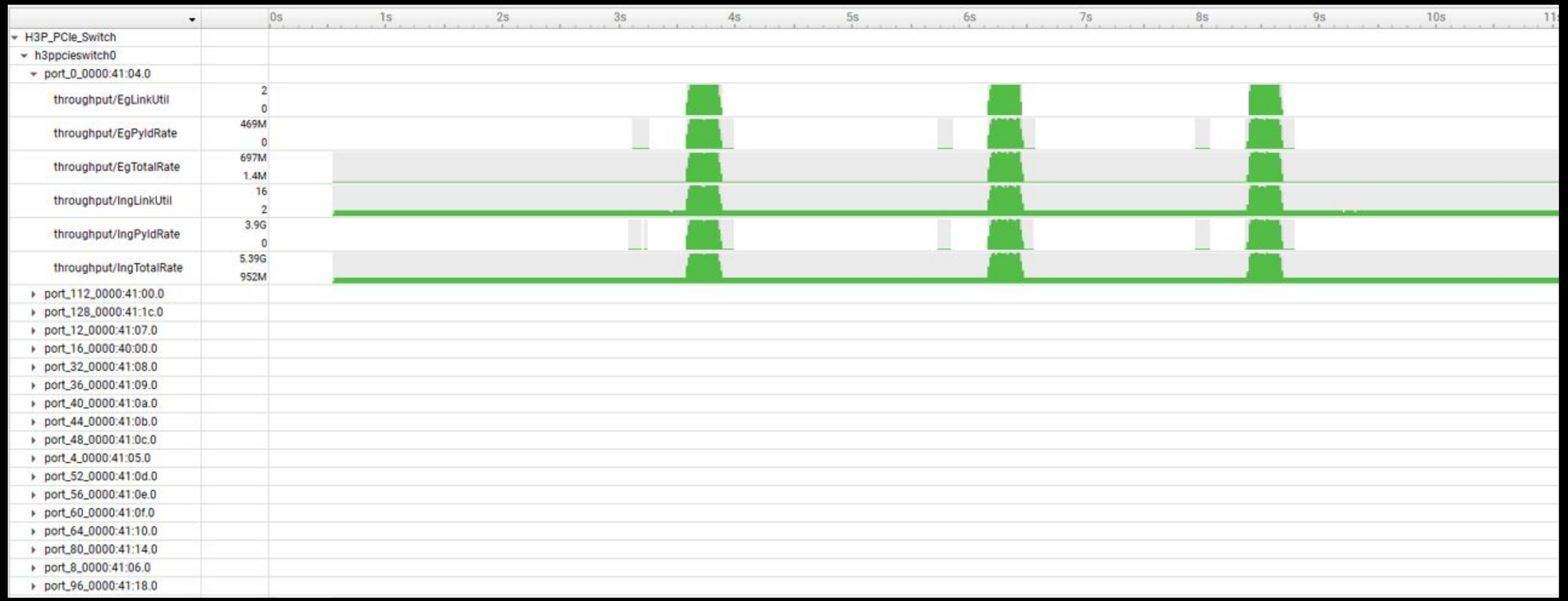
- 2 CPU Upstreams
- 2 NVIDIA GPUs
- 4 Networkd Cards
- 48 NVMe SSDs

SCADA and CLI Binary by H3 Platform



Nsight Integration for GPU, NVMe, NIC Profiling

Show the performance of all devices attached to the PCIe switches (will attached all devices and show the data in a later version)



Demo Video-- 5 Minutes

Time	Topics	Detailed description
0:00–0:25	Demo purpose	Introduce the H3 system and the GPU storage workload. Explain that the demo follows application activity and PCIe traffic to understand where the workload spends time.
0:25–0:55	System topology and test configuration	Show the H3 topology view. Identify the GPU, selected SSDs, switch ports, and shared uplinks. Display the actual hardware configuration, I/O size, access pattern, and cache conditions.
0:55–1:20	Nsight collection setup	Show the H3 plugin in the available-plugin list and the capture command. Explain that the report collects application traces and switch telemetry. Cut directly to the completed report.
1:20–1:50	Timeline navigation	Open the saved Nsight report. Expand application markers, CUDA activity, and H3 switch tracks. Select a steady I/O interval, excluding initialization and warm-up.
1:50–2:40	PCIe performance information	Highlight the mapped GPU and SSD ports. Explain throughput, traffic direction, units, and utilization where available. Compare activity across the path to identify a possible limiting stage.

Demo Video-- 5 Minutes

Time	Topics	Detailed description
2:40–3:25	GPU and storage correlation	Compare the selected PCIe interval with GPU execution. Explain whether data access overlaps compute or precedes it. Show SCADA server counters or latency distributions if the deployed software exposes them.
3:25–4:15	Controlled performance comparison	Compare low and higher request-concurrency runs while keeping other conditions fixed. Show measured IOPS, useful bandwidth, workload duration, and latency where available. Explain whether additional concurrency improves throughput or increases waiting. (4K vs 512B)
4:15–5:00	Findings and next steps	Summarize the observed behavior and the suspected constraint. State the next measurement needed to confirm it. Remind attendees to compare the same interval and change one parameter at a time.